SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
   Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.
5. Given:
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   $y = w \times x$
   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I __S.Venkateswarlu (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:S.Venkateswarlu

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Coin Toss (MLE)

Given

- Number of tosses (n) = 10
- Outcomes = H H T H H H T H H T
- Number of Heads (H) = 7
- Number of Tails (T) = 3

Formula

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is

$$\boxed{\hat{p} = 0.7}$$

Interpretation

The estimated probability of getting a Head is **70%**.

2. Student Pass Probability (MLE)

Given

- Total students = 50
- Passed = 42

Formula

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Observations}}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer

$$\boxed{\hat{p} = 0.84}$$

Interpretation

The estimated probability that a student passes the exam is **84%**.

3. Defective Bulbs (MLE)

Given

- Total bulbs = 100

- Defective bulbs = 8

Formula

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer

$$\boxed{\hat{p} = 0.08}$$

Interpretation

The estimated probability that a bulb is defective is **8%**.

4. Biased Die (MLE)

Given

- Total rolls = 60
- Number of times 6 appears = 18

Formula

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.30$$

Answer

$$\boxed{\hat{p} = 0.30}$$

Interpretation

The estimated probability of rolling a **6** is **30%**.

5. Gradient Descent Weight Update

Given

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Gradient = 4
- Learning Rate (α) = 0.01

Step 1: Predicted Output

Formula

$$y = w \times x$$

Calculation

$$y = 2 \times 4 = 8$$

Answer

$$\boxed{\text{Predicted Output} = 8}$$

Step 2: Error
Formula

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation

$$10 - 8 = 2$$

Answer

$$\boxed{\text{Error} = 2}$$

Step 3: Updated Weight
Formula

$$w_{\text{new}} = w - \alpha \times \text{Gradient}$$

Calculation

$$\begin{aligned} w_{\text{new}} &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96 \end{aligned}$$

Answer

$$\boxed{w_{\text{new}} = 1.96}$$

Interpretation
The weight is updated from **2** to **1.96**, helping the model reduce prediction error during the learning process.